The Halvorsen–Skogestad conjecture in the Underwood model: the generalized Petlyuk arrangement minimizes total vapor among networks of two-section distillation columns

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Abstract. Halvorsen and Skogestad (Ind. Eng. Chem. Res. 2003) conjectured that the generalized Petlyuk arrangement (fully thermally coupled, FTC) needs less vapor than any other adiabatic distillation arrangement without internal heat exchange, for the separation of an ideal N-component feed into pure products. They did not define the class of competitors, and they proved the minimum-vapor formula of the Petlyuk arrangement itself (the highest peak of the $V_{\min}$ diagram) only for the extreme peaks. We take as competitors all networks of two-section columns with interval streams, side draws, thermal couplings or heat exchangers on any link and heaters or coolers at any node; they contain Agrawal's regular-column configurations (152 for $N = 4$, 6128 for $N = 5$). Each column obeys the infinite-stage Underwood model of Tumbalam Gooty, Agrawal and Tawarmalani, without reverse flows. We prove: (i) in the strict form of the model, in which every column with two or more (nominally) distributing components runs exactly at its Underwood vapor flow, every network needs a total vapor generation of at least $\max_j H_j$, the reboiler-referenced highest peak, for every N; the FTC attains this value, so the conjecture holds in this class and model; (ii) in the permissive form used for global optimization, which also lets a column carry excess vapor, the same bound holds for the regular-column configurations with $N \le 4$, for every network and every N when the highest peak is the first or the last one, for every regular-column configuration and every N whose mixed sub-network for the highest peak has no merge node (under a non-degeneracy assumption), and for the FTC itself for every N, which proves the Halvorsen–Skogestad formula for the minimum vapor of the generalized Petlyuk arrangement in this model (without reverse flows). Conversely, the permissive bound fails for $N = 5$: the independent verification found a five-component configuration whose permissive minimum lies below $\max_j H_j$ (relative gap $5 \cdot 10^{-4}$, existence of the exact point proved by an interval Krawczyk test). It uses excess vapor in a column in which two components distribute, which no infinite-stage column with pure products can carry, so it is an artifact of the permissive formulation, and the restriction to the strict model is essential. The proofs combine an exact cut identity for the Underwood flux $\Lambda_s(\theta_j) = V_s - \sum_p \alpha_p w_{s,p}/(\alpha_p - \theta_j)$ with a sign rule for each column and, for one four-component family, an ordering of recoveries. Columns with several feed stages, stream splitters, recycles and reverse flows are outside the class; the conjecture remains open there, and at degenerate operating points (a nominally distributing component absent from one product) the strict model is more restrictive than infinite-stage physics (Section 2.2).

1 The question

I. J. Halvorsen and S. Skogestad, *Minimum energy consumption in multicomponent distillation. 3. More than three products and generalized Petlyuk arrangements*, Ind. Eng. Chem. Res. 42 (2003) 616–629 [1]. The abstract states:

> *"We also conjecture that the minimum energy requirement for the generalized extended Petlyuk arrangement is lower than the minimum-energy requirement for any distillation configuration when we consider conventional adiabatic sections and no internal heat exchange."*

Section 6.3 (p. 626) makes the assumptions explicit and records the status:

> *"We assume constant relative volatilities, constant molar flows, constant pressure, and no internal heat integration. Then the generalized adiabatic Petlyuk arrangement has the lowest need for vaporization compared to any other adiabatic distillation arrangement for separation of an arbitrary feed mixture into its pure components. We have not presented a complete proof, so the above conclusion is a conjecture."*

The number of stages is infinite; an adiabatic section is "a column section with constant molar flows and no heat exchange along the section". Even the minimum vapor of the Petlyuk arrangement itself was left partly open (Section 5.3, p. 624):

> *"We have not carried out a detailed proof for the general N-component M-product case for other than the far left and right peaks."*

For three components, Fidkowski and Królikowski [2] compared the Petlyuk column with side strippers, side rectifiers and the direct and indirect sequences in closed form. Global MINLP studies of the Agrawal group find the FTC value optimal on the instances they solve [3] [4], and later papers state the property as known; we found no proof (novelty check in `references.md`). Counterexamples are known only when condensers and reboilers exchange heat [5], which the conjecture excludes.

Halvorsen and Skogestad do not define "any adiabatic distillation arrangement". A theorem must fix the class, and the model that computes the minimum vapor of each member. We use networks of two-section columns with interval streams (Section 2), which contain the regular-column configurations of Agrawal and co-workers [6] [3], and the infinite-stage Underwood model of Tumbalam Gooty, Agrawal and Tawarmalani [3]. Our results in that setting:

Theorem 1 (main results). Let $\alpha_1 > \cdots > \alpha_N > 0$, $z_p > 0$, $\sum z_p = 1$ and $q \in \mathbb{R}$. Let $\theta_j \in (\alpha_{j+1}, \alpha_j)$ be the roots of $\sum_p \alpha_p z_p/(\alpha_p - \theta) = 1 - q$ and $H_j = \sum_{p>j} \alpha_p z_p/(\theta_j - \alpha_p)$ $(1 \le j \le N-1)$. Let R denote the total vapor generated by all heat exchangers (reboilers) of an arrangement.

(1) (Strict model, every N.) At every feasible operating point of every two-section column network (Definition 2), in particular of every regular-column configuration, $R \ge \max_j H_j$.

(2) (Permissive model.) The same inequality holds at every feasible point (a) for every regular-column configuration when $N \le 4$; (b) for every network and every N, for $j = 1$ and $j = N-1$; (c) for the FTC configuration and every N, for all j; (d) for every regular-column configuration and every N, for every j such that no mixed stream (one containing j and $j+1$) is produced by two columns, under a non-degeneracy assumption (N).

(3) (Achievability.) The FTC configuration has an operating point, feasible in both models, with $R = \max_j H_j$.

Consequently the generalized Petlyuk (FTC) arrangement minimizes total vapor generation among all two-section column networks for every N in the strict model, and among all regular-column configurations for $N \le 4$ in the permissive model; its minimum is exactly $V_{\mathrm{FTC}} := \max_j H_j$, the highest peak of the $V_{\min}$ diagram referred to the reboiler, for every N in both models.

Part 1 is Theorem 11, part 2 is Theorems 15, 19, 23 and 18, part 3 is Theorem 24. Part 2 does not extend to all configurations: for $N = 5$ the permissive model has a configuration and data with minimum $R < \max_j H_j$ (Section 6.3, found during the independent verification). With $P_j^{\mathrm{T}} = \sum_{p \le j} \alpha_p z_p/(\alpha_p - \theta_j)$ (the j-th peak of the $V_{\min}$ diagram, top vapor of a two-product column performing the sharp split $\{1..j\}|\{j{+}1..N\}$), the feed equation gives $H_j = P_j^{\mathrm{T}} - (1-q)$, so V_{FTC} is the value of [1], eq. (8), referred to the reboiler. The two standing hypotheses of the model, stated in Section 2.2, are: no reverse flows (net component flows of fixed sign in every section), and a single feed with pure, saturated-liquid products. The argument is entirely analytic; the scripts of Section 8 check identities, search for counterexamples and confirm the case analysis, but no step of the proofs depends on floating-point computation.

2 Setting

2.1 Networks of two-section columns

A *stream* is an interval $[i, m] = \{i, \dots, m\}$ of components ($i \le m$); $[p, p]$ is the pure product p and $[1, N]$ the feed.

Definition 2 (two-section column network)**.** A network consists of the N pure-product nodes and a finite set of *columns*. A column c has a label $[i, m]$ ($i < m$) and a split $[i, k]/[l, m]$ with $i \le k < m$, $i < l \le m$ and $l \le k + 1$ (components $l, \dots, k$ distribute). It has one feed stage, between a top and a bottom adiabatic section of infinitely many stages. Its top section delivers the distillate $\mathrm{dist}(c) = [i, k]$ to one node with that label (a column labelled $[i, k]$, or the pure product i if $k = i$), and its bottom section delivers the residue $\mathrm{res}(c) = [l, m]$ to one node labelled $[l, m]$. Exactly one column, the *feed column*, is labelled $[1, N]$ and receives the process feed; every other column receives at least one stream. A node may receive any number of streams (from several producers, as a distillate or as a residue), and several columns may carry the same label. Each link is a thermal coupling (the vapor and liquid of the producing section are exchanged directly at the receiving node) or passes through a heat exchanger, and every node may carry an external heat duty: vapor generated (a reboiler or heater, counted in R) or vapor condensed (a condenser or cooler). Pure products are withdrawn as saturated liquid.

Labels shrink strictly along links, so a network is acyclic and every column is reached from the feed column. Not covered: columns with several feed stages (intermediate sections between feeds), stream splitters that divide one section product among several columns, recycles, non-interval intermediate streams, splits in which a product keeps every component of the column's feed (excluded by $k < m$ and $l > i$), streams delivered to a node whose label differs from the stream's label, and the reverse flows excluded in

Section 2.2.

Regular-column configurations [6] [3] are the networks with at most one column per label, in which the distillate of $[i,m]$ is $[i,k]$ for the largest $k<m$ such that a column $[i,k]$ exists (or $k=i$), the residue is $[l,m]$ for the smallest such $l>i$, a stream produced by one column is transferred by a thermal coupling or by a condenser (distillate; TG: a partial condenser removing the reflux, the stream leaving as saturated vapor) or a reboiler (residue; the stream leaving as saturated liquid), a stream produced by two columns is a side draw (thermal couplings), and a pure product is withdrawn through a total condenser or a reboiler, or as a liquid side draw between the two sections that produce it. There are 8, 152 and 6128 such configurations for $N=3,4,5$ (transfer choices included). The *FTC* (generalized Petlyuk) configuration contains every mixture stream, all links being thermal couplings or side draws; it has one reboiler and one condenser.

2.2 The Underwood model

Constant relative volatilities, constant molar overflow, constant pressure, infinitely many stages in every section. For a column c with split $[i,k]/[l,m]$: net upward component flows $d_p\ge 0$ in the top section ($i\le p\le k$), net downward flows $b_p\ge 0$ in the bottom section ($l\le p\le m$), net feed $f_p=d_p[p\le k]+b_p[p\ge l]$ (the feed z for the feed column, otherwise the sum of the incoming flows), vapor flows $V_T,V_B\ge 0$ with $V_T-V_B=U_c$, where U_c is the net vapor entering the feed stage: $1-q$ for the feed column; otherwise the vapor brought by the incoming distillate links, minus the vapor drawn by thermally coupled residue links, plus the vapor generated minus the vapor condensed at the node. In a regular-column configuration with TG's conventions, a thermally coupled or side-draw distillate link brings V_T(producer), a condensed one $D=V_T-L_T$, a coupled residue link draws V_B(producer) and a reboiled one draws nothing. Put

$$g_T(x)=\sum_{p\le k}\frac{\alpha_p d_p}{\alpha_p-x}\,,\qquad g_B(x)=-\sum_{p\ge l}\frac{\alpha_p b_p}{\alpha_p-x}\,.$$

For every $r\in\{l-1,\dots,k\}$ there is a root $\theta_{c,r}\in[\alpha_{r+1},\alpha_r]$ of the feed equation $\sum_p \alpha_p f_p/(\alpha_p-\theta_{c,r})=U_c$, and a threshold vapor Y_T (with $Y_B=Y_T-U_c$) such that

$$\begin{gathered} g_T(\theta_{c,r})\le Y_T\quad (r=l-1,\ r=k),\qquad g_T(\theta_{c,r})=Y_T\quad (l\le r\le k-1),\\ 0\le Y_T\le V_T,\qquad 0\le Y_B\le V_B. \end{gathered}\tag{3}$$

This is TG's model [3] (constraints (A18)–(A23) of the arXiv version, including the equalities at the interior roots which, TG note, are needed for the correct degrees of freedom); we call it the *permissive* model. (Nallasivam et al. [4] use a closely related formulation: an inequality $V_T\ge g_T(\theta)$ at every root together with explicit recovery-ordering constraints, instead of TG's equalities at the interior roots. The results below are proved for TG's formulation.) The *strict* model adds $Y_T=V_T$ whenever the column has an interior root ($l\le k-1$): the column then runs exactly at the Underwood vapor of each of its interior roots, as an infinite-stage column does whenever the components $l,\dots,k$ actually distribute (then these roots are active): "we may apply the common roots being in the range of volatilities for the components distributed to both ends. We denote these roots active roots. If we have any active roots then $V=V_{\min}$" (Halvorsen [7], §4.2.1; see also §3.2.3–3.2.4). The permissive model also admits columns carrying vapor above their threshold ("a column may be operated above

its threshold vapor requirement to supply the vapor to another column" [3]), physically a vapor bypass or a finite-stage effect. Strict-feasible points are permissive-feasible.

Degenerate operating points (added by the verifier). Both models fix the split of each column through the network, so the strict model imposes $Y_T = V_T$ whenever two or more components are *nominally* distributing, whether or not they appear in both products. At a point where every component $l, \dots, k$ has a positive flow in both products the interior roots are active and the strict model is exactly the infinite-stage behaviour. At a degenerate point, where a nominally distributing component is absent from one product (e.g. $d_C = 0$ in a split ABCD→ABC/BCD), an infinite-stage column is governed by its actual distribution: Halvorsen counts a component as distributing only if it "appears in both products, or is exactly at the limit of becoming distributing" ([7], §3.2.4), so with fewer than two distributing components no root is active and the column may physically carry vapor above its Underwood value, which the strict model forbids; TG's equalities at the nominal interior roots are then not physically required either. Relabelling the split does not help in general, because the delivered stream would no longer carry the label of its receiver (Definition 2). Such points are not covered by Theorem 11; the theorems are statements about the model as defined. (The results that use only the sign rule, Lemma 8, rest on the inequality $V_T \geq g_T(\theta)$ at a common root, which holds at any infinite-stage operating point with nonnegative flows; we expect, but have not written out, their extension to degenerate points.)

At the roots, $g_T - g_B = U_c$, hence $g_B(\theta_{c,r}) \leq Y_B$ for $r = l-1, k$ with equality for interior r.

Standing hypotheses. (H1) No reverse flows: $d_p \geq 0$ and $b_p \geq 0$ in every section. This is part of the model of [3] [4]: the conditions (3) express Underwood's root interlacing for top sections with $w \geq 0$ and bottom sections with $w \leq 0$, and with reverse flows the roots can leave their intervals (Halvorsen [7], §4.3.3–4.3.4). Reverse flows do occur in directly coupled columns operated away from the preferred split, e.g. in a Petlyuk prefractionator whose top liquid carries a component that the top section would otherwise remove (*ibid.*, §4.3.3), so (H1) is a genuine restriction of the operating points, not only of the configurations. The cut identity below needs no sign assumption, but the column lemmas use (H1) through the monotonicity of the Underwood functions; whether reverse flows could lower R is not addressed here. (H2) A single feed, and pure products withdrawn as saturated liquid.

The objective is R, the vapor generated by all heat exchangers (reboilers on products and on transfer streams, heaters at nodes).

2.3 Peaks and regions

Fix $j \in \{1, \dots, N-1\}$ and write $\theta = \theta_j$. A stream, column or node with label $[i, m]$ is *light* if $m \leq j$, *heavy* if $i \geq j+1$, *mixed* if $i \leq j < m$. Producers of a mixed column are mixed (their labels contain its label); the distillate of a mixed column is light or mixed (it contains $i \leq j$), its residue mixed or heavy; the sections of a light (heavy) column join light (heavy) nodes. A *light exit* is the top section of a mixed column whose distillate is light; a *heavy exit* is the bottom section of a mixed column whose residue is heavy.

3 The Underwood flux and the cut identity

For a section s with vapor flow V_s and upward net component flows w_s ($w = d$ in a top section, $w = -b$ in a bottom section), define the *Underwood flux*

$$\Lambda_s(x) = V_s - \sum_p \frac{\alpha_p w_{s,p}}{\alpha_p - x}\,.$$

For a column c: $\Lambda_{c,T} = V_T - g_T$, $\Lambda_{c,B} = V_B - g_B$, and its *feed function* $\Lambda_{c,F}(x) = U_c - \sum_p \alpha_p f_p/(\alpha_p - x)$ satisfies $\Lambda_{c,T} - \Lambda_{c,B} = \Lambda_{c,F}$.

Lemma 4 (monotonicity). On any interval free of poles, $\Lambda_{c,T}$ and $\Lambda_{c,F}$ are nonincreasing and $\Lambda_{c,B}$ is nondecreasing; $\Lambda_{c,F}$ is strictly decreasing if some $f_p > 0$.

Proof. The derivatives are $-\sum \alpha_p d_p/(\alpha_p - x)^2$, $-\sum \alpha_p f_p/(\alpha_p - x)^2$ and $+\sum \alpha_p b_p/(\alpha_p - x)^2$. □

3.1 Nodes and the identity

The *nodes* of a network are its columns and its pure products: a column node gathers the feed stage of the column, the heat exchangers on its incoming links and its external duty; a product node gathers the withdrawal point and its heat exchanger (or the side draw). Every section joins the node of its column to the node that receives its distillate (top section) or its residue (bottom section). We orient sections upward and write $s\colon X \to Y$ when the bottom end of s is in X and its top end in Y.

Proposition 5 (cut identity). Let U be a set of nodes containing every light pure product and no heavy pure product, U^c its complement. Then

$$R - H_j = R(U) + Q(U^c) + \sum_{s\colon U^c \to U} \Lambda_s(\theta_j) - \sum_{s\colon U \to U^c} \Lambda_s(\theta_j), \tag{6}$$

where $R(U)$ is the reboiler duty (vapor generated) located in U and $Q(U^c) \geq 0$ the condenser duty (vapor condensed) located in U^c.

Proof. $\Lambda_s(x)$ is linear in (V_s, w_s). At a node, constant molar overflow and the absence of heat exchange give conservation of the total vapor flow and of the net component flows, except for: the feed, which injects vapor $1 - q$ and components z; a heat exchanger, which converts liquid into vapor (reboiler, duty Q_r) or vapor into liquid (condenser, duty Q_c); a liquid product withdrawal of z'_p of component p. A section whose top end is at the node brings the flux Λ_s into it; a section whose bottom end is at the node takes Λ_s out of it (its vapor leaves upward, its liquid arrives, its net flow w_s leaves upward). Hence, at every node,

$$\sum_{s \text{ ending at the node}} \Lambda_s(x) - \sum_{s \text{ starting at the node}} \Lambda_s(x) = Q_c - Q_r - \Lambda^{\text{feed}}(x) - \frac{\alpha_p z'_p}{\alpha_p - x}\,,$$

with $\Lambda^{\text{feed}}(x) = (1 - q) - \sum_p \alpha_p z_p/(\alpha_p - x)$ at the feed column only and the last term at product nodes only. (Agent C verified this identity in exact rational arithmetic on all configurations for $N \leq 5$, with arbitrary junction duties.) At $x = \theta_j$, $\Lambda^{\text{feed}} = 0$ by the definition of θ_j. Sum the balances over the nodes of U^c: sections inside U^c cancel, every heavy product withdraws $\alpha_p z_p/(\theta_j - \alpha_p)$, in total H_j, and no light product lies in U^c. This gives $R(U^c) = H_j + Q(U^c) + \sum_{U^c \to U} \Lambda_s - \sum_{U \to U^c} \Lambda_s$; add $R(U)$. □

The canonical choice $U =$ light streams gives

$$R - H_j = R(\text{light}) + Q(\text{mixed}) + Q(\text{heavy}) + \sum_{\text{light exits}} \Lambda_s(\theta_j), \tag{7}$$

and $U =$ light $\cup$ mixed streams gives $R - H_j = R(\text{light}) + R(\text{mixed}) + Q(\text{heavy}) + \sum_{\text{heavy exits}} \Lambda_s(\theta_j)$. So $R \geq H_j$ as soon as the right-hand side of (6) is nonnegative for one admissible U; conversely, the inequality is tight on open sets of parameters for many configurations (Agent C's computations), so no strict inequality can be expected.

4 Column lemmas

For a mixed column c put $A_c = \Lambda_{c,T}(\theta_j)$, $B_c = \Lambda_{c,B}(\theta_j)$ and $a_c = \Lambda_{c,F}(\theta_j) = A_c - B_c$.

Lemma 8 (sign rule)**.** For every mixed column, in both models: $A_c \geq \min(0, a_c)$, equivalently $B_c \geq \min(0, -a_c)$. In particular A_c and B_c are never both negative, $a_c \geq 0 \Rightarrow A_c \geq 0$ and $a_c \leq 0 \Rightarrow B_c \geq 0$. Moreover, if $k < j$ then $A_c \geq 0$, and if $l > j + 1$ then $B_c \geq 0$.

Proof. Let the split be $[i, k]/[l, m]$; (3) and $Y \leq V$ give $\Lambda_{c,T}(\theta_{c,r}) \geq 0$ and $\Lambda_{c,B}(\theta_{c,r}) \geq 0$ for $r \in \{l - 1, \ldots, k\}$. The values are finite: if $\theta_{c,r}$ is an endpoint $\alpha_{r'}$, the feed equation forces $f_{r'} = 0$, hence $d_{r'} = b_{r'} = 0$.

Case $l - 1 \leq j \leq k$. If all flows vanish the claim is trivial. Otherwise $\Lambda_{c,F}$ is strictly decreasing on $[\alpha_{j+1}, \alpha_j]$ (where finite) and vanishes at $\theta_{c,j}$. If $a_c \geq 0$ then $\theta_j \leq \theta_{c,j}$, and $A_c \geq \Lambda_{c,T}(\theta_{c,j}) \geq 0$ by Lemma 4. If $a_c \leq 0$ then $\theta_j \geq \theta_{c,j}$ and $B_c \geq \Lambda_{c,B}(\theta_{c,j}) \geq 0$, so $A_c = a_c + B_c \geq a_c$.

Case $k < j$. The top section contains only components $p \leq k$, so $\Lambda_{c,T}$ has no pole in $(-\infty, \theta_{c,k}]$; since $\theta_j < \alpha_j \leq \alpha_{k+1} \leq \theta_{c,k}$, $A_c \geq \Lambda_{c,T}(\theta_{c,k}) \geq 0$; and if $a_c \leq 0$, $B_c = A_c - a_c \geq 0$. *Case* $l > j + 1$: symmetric, with $\theta_{c,l-1} < \alpha_{j+1} < \theta_j$ and $\Lambda_{c,B}$ nondecreasing on $[\theta_{c,l-1}, \infty)$. □

Lemma 9 (strict model: interior split)**.** In the strict model, if both the distillate and the residue of a mixed column are mixed ($l \leq j \leq k - 1$), then A_c and B_c are never both positive.

Proof. j is an interior root of c, so $V_T = Y_T = g_T(\theta_{c,j})$: $\Lambda_{c,T}(\theta_{c,j}) = \Lambda_{c,B}(\theta_{c,j}) = 0$. If $\theta_j \leq \theta_{c,j}$ then $B_c \leq 0$; if $\theta_j \geq \theta_{c,j}$ then $A_c \leq 0$ (Lemma 4). □

For the feed column $a_{[1,N]} = \Lambda^{\text{feed}}(\theta_j) = 0$, so $A = B \geq 0$ by Lemma 8. *Propagation.* By the node balance of Proposition 5, a column c other than the feed column has

$$a_c = \sum_{P:\ \text{dist}(P)=c} A_P - \sum_{P':\ \text{res}(P')=c} B_{P'} + \rho_c - \kappa_c, \tag{10}$$

where $\rho_c \geq 0$ is the vapor generated and $\kappa_c \geq 0$ the vapor condensed at the node of c (in TG's convention: $\kappa_c = L_T(P)$ for a condensed distillate link, $\rho_c = V_B(P')$ for a reboiled residue link, 0 for couplings and side draws).

5 The strict model: all N

Theorem 11. In the strict model, every feasible operating point of every two-section column network satisfies $R \geq H_j$ for every j.

The argument is due to the computation agent (Agent C) and was checked by the theory agent; a standalone machine check of its combinatorial core is described in Section 8.

Proof. Fix j and an operating point. Build the *sign graph*: for every section $s\colon x \to y$, an arc $x \to y$ if $\Lambda_s(\theta_j) > 0$ and an arc $y \to x$ if $\Lambda_s(\theta_j) < 0$. Let U be the set of nodes reachable from the light nodes (light nodes included). If U contains no heavy node, then every section $s\colon U^c \to U$ has $\Lambda_s \geq 0$ (otherwise its bottom end would be reachable) and every section $s\colon U \to U^c$ has $\Lambda_s \leq 0$; so all terms of (6) are nonnegative and $R \geq H_j$.

We show by induction on the length of a shortest path from the light nodes that every reached mixed node y satisfies (i) $B_y \geq 0$ and (ii) $A_y \leq 0$ if $\mathrm{dist}(y)$ is mixed, and that no heavy node is reached. The sections incident to a mixed column node y are its top section (to the node receiving $\mathrm{dist}(y)$), its bottom section (from the node receiving $\mathrm{res}(y)$) and the sections of its producers ending at y; we write $\mathrm{dist}(y)$, $\mathrm{res}(y)$ also for these nodes. Let y be reached by an arc from an already reached node z.

- $z = \mathrm{dist}(y)$ (light or mixed), arc down the top section of y: then $A_y < 0$, so $B_y \geq 0$ by Lemma 8; (i) and (ii) hold.
- $z = \mathrm{res}(y)$, arc up the bottom section of y: then $B_y > 0$. The node z is mixed (a residue of a mixed column is not light, and no heavy node has been reached). If $\mathrm{dist}(y)$ is mixed, both products of y are mixed and Lemma 9 gives $A_y \leq 0$. So (i) and (ii) hold.
- z a producer of y (necessarily mixed): an arc up the top section of z needs $A_z > 0$, excluded by (ii) for z since $\mathrm{dist}(z) = y$ is mixed; an arc down the bottom section of z needs $B_z < 0$, excluded by (i) for z.

A heavy node y could only be entered from a mixed column c with $\mathrm{res}(c) = y$, down the bottom section of c, which needs $B_c < 0$, against (i); no section joins a light node to a heavy node. Hence U contains no heavy node. □

Corollary 12 (necessary condition for a tie; Agent C)**.** In the strict model, every feasible point of every network satisfies $R - H_j \geq R(\mathrm{light}) + Q(\mathrm{heavy})$ for every j, where $R(\mathrm{light})$ is the vapor generated at nodes with labels inside $[1, j]$ and $Q(\mathrm{heavy})$ the vapor condensed at nodes with labels inside $[j+1, N]$. Hence, if H_{j^*} is the strict maximum of the peaks, a network can attain V_{FTC} only if it has no reboiler or heater with positive duty at a node inside $[1, j^*]$ and no condenser or cooler with positive duty at a node inside $[j^*+1, N]$.

Proof. In the proof of Theorem 11, U contains the light nodes, U^c the heavy ones, and all flux terms of (6) are nonnegative; so $R - H_j \geq R(U) + Q(U^c) \geq R(\mathrm{light}) + Q(\mathrm{heavy})$. □

In Agent C's runs ($N = 4$, $q = 1$), 2717 of 2720 observed ties satisfy this condition; the three exceptions are near-ties (relative gap $\sim 10^{-9}$) caused by trace components.

Remark 13. The proof uses only Lemmas 8–9 column by column, the interval labels and the identity (6). Heat duties never create arcs and enter (6) with nonnegative signs, which is why the theorem holds for the whole class of Definition 2: any thermal state of the transferred streams, heaters or coolers at any node, side draws merging any number of links, several columns with the same label, any q, and no regularity rule. Hypothesis (H1) enters through Lemma 4. ⌟

6 The permissive model

In the permissive model Lemma 9 fails: a column with an interior root may run above its threshold, with $A_c = B_c = V_T - Y_T > 0$. The reachability argument then leaves possible "bad paths"; Agent C's exact enumeration finds none for $N = 3$, exactly one family for $N = 4$ (family K below), 70 paths in 48 basic configurations for $N = 5$ and 5795 in 2300 for $N = 6$.

6.1 Extreme peaks and paths

Lemma 14 (path)**.** In a network in which every mixed column has at most one mixed child (the node receiving its distillate or its residue), $R \geq H_j$ in both models.

Proof. Every mixed column is reached from the feed column through mixed producers, and each has at most one mixed child; hence the mixed columns form a path $c_0 \to c_1 \to \cdots \to c_n$ (c_0 the feed column), and c_{t+1} has the single producer c_t. Write a_t, A_t, B_t for c_t, $\rho_t, \kappa_t \geq 0$ for the vapor generated and condensed at the node of c_{t+1}, T_t for the sum of A over the light exits among $c_t, \ldots, c_n$ and $K_t = \sum_{t' \geq t} \kappa_{t'}$. We claim $T_t \geq \min(0, a_t) - K_t$. For $t = n$: $T_n = A_n \geq \min(0, a_n)$ (Lemma 8; the top section of c_n is a light exit). If $c_{t+1} = \mathrm{dist}(c_t)$: $T_t = T_{t+1}$ and $a_{t+1} = A_t + \rho_t - \kappa_t \geq \min(0, a_t) - \kappa_t$. If $c_{t+1} = \mathrm{res}(c_t)$, the top of c_t is a light exit and $a_{t+1} = -B_t + \rho_t - \kappa_t$; then $T_t = A_t + T_{t+1} \geq A_t + \min(0, a_{t+1}) - K_{t+1}$, which is $\geq A_t - K_{t+1} \geq \min(0, a_t) - K_t$ if $a_{t+1} \geq 0$, and $\geq A_t - B_t - \kappa_t - K_{t+1} = a_t - K_t$ otherwise. With $a_0 = 0$: $\sum_{\text{light exits}} \Lambda_s = T_0 \geq -K_0 \geq -Q(\text{mixed})$, and (7) gives $R \geq H_j$. □

Theorem 15 (extreme peaks)**.** In both models, for every N and every two-section column network, $R \geq H_1$ and $R \geq H_{N-1}$.

Proof. For $j = 1$ the mixed columns are labelled $[1, m]$, $m \geq 2$; their residues are heavy, so each has at most one mixed child (its distillate) and Lemma 14 applies. For $j = N - 1$ the mixed columns are labelled $[i, N]$ and their distillates are light. □

For $N = 3$ both peaks are extreme: the conjecture holds in both models for all eight ternary configurations and every q. For any N, the conjecture holds in both models whenever the highest peak is H_1 or H_{N-1}.

6.2 Recovery ordering and merge-free mixed networks

The results of this subsection and their proofs are due to the general-N agent (Agent G, `general/general-N.md`); they hold for regular-column configurations under the non-degeneracy assumption

> (N) every column receives a positive net flow of every component of its label.

Lemma 16 (recovery ordering). Let c be a column with split $[i,k]/[l,m]$, at a point satisfying (3), with $f_r > 0$ for all $r \in [i,m]$. Put $d_r := f_r$ for $r < l$ and $d_r := 0$ for $r > k$. Then $d_p/f_p \geq d_q/f_q$ for all $p < q$ in $[i,m]$; equivalently $d_p b_q \geq d_q b_p$.

Proof. Only $l \leq p < q \leq k$ needs proof. Let $D = Y_T - g_T$, $\tilde{B}(x) = Y_B + \sum_{r\geq l} \alpha_r b_r/(\alpha_r - x)$ and $F = D - \tilde{B} = U_c - \sum_r \alpha_r f_r/(\alpha_r - x)$. By (3), $D(\theta_{c,r}) = \tilde{B}(\theta_{c,r})$, equal to 0 for interior r and ≥ 0 for $r \in \{l-1, k\}$. Suppose $d_p/f_p < \lambda < d_q/f_q$ with λ different from every d_s/f_s, so $0 < \lambda < 1$, and put $H = (1-\lambda)D + \lambda\tilde{B} = D - \lambda F = K_0 - \sum_{s=i}^{m} \alpha_s c_s/(\alpha_s - x)$ with $c_s = d_s - \lambda f_s \neq 0$. The signs σ_s of c_s are $+$ for $s < l$, $-$ for $s > k$, $\sigma_p = -$, $\sigma_q = +$; the sequence $\sigma_i, \dots, \sigma_m$ has $C \geq 3$ sign changes, all between $s = l-1$ and $s = k+1$. Write $H = P(x)/\prod_s(\alpha_s - x)$, $\deg P \leq n := m - i + 1$, $P \not\equiv 0$. Near α_s, $H \approx -\alpha_s c_s/(\alpha_s - x)$, so on $I_r = (\alpha_{r+1}, \alpha_r)$, $i \leq r < m$, H has sign σ_{r+1} at the left end and $-\sigma_r$ at the right end. If $\sigma_r = \sigma_{r+1}$, P has a zero in I_r. If $\sigma_r \neq \sigma_{r+1}$, then $l - 1 \leq r \leq k$, both ends of I_r have the same sign, and $H(\theta_{c,r}) = 0$ (r interior) or $H(\theta_{c,r}) \geq 0$ with both ends negative ($r = l-1$: ends $\sigma_l = -$ and $-\sigma_{l-1} = -$; $r = k$: ends $\sigma_{k+1} = -$ and $-\sigma_k = -$); either way P has at least two zeros in I_r, counted with multiplicity. In total P has at least $(n - 1 - C) + 2C \geq n + 2$ zeros, a contradiction. □

Lemma 21 below is the case $(p,q) = (\mathrm{B}, \mathrm{C})$ of the feed column of K, with a direct proof that does not need (N).

Light and heavy parts. A function $y(x) = U - \sum_p \alpha_p w_p/(\alpha_p - x)$ with $w \geq 0$ (nonincreasing on I_j) is written $y = U - G_y + K_y$ with $G_y(x) = \sum_{p\leq j} \alpha_p w_p/(\alpha_p - x)$ (nondecreasing on I_j) and $K_y(x) = \sum_{p>j} \alpha_p w_p/(x - \alpha_p)$ (nonincreasing). This applies to the feed function $a_c := \Lambda_{c,F}$, to $\Lambda_{c,T}$, to $-\Lambda_{c,B}$ and to the fluxes carried by the links into a mixed column: $y_e = \Lambda_{P,T} - \kappa_e$ for a distillate link from P, $y_e = -\Lambda_{P',B} + \rho_e$ for a residue link from P' (κ_e, ρ_e the duties of a condenser or reboiler on the link), so that $a_c = \sum_e y_e$ as functions on I_j (cf. (10)). Under (N) each of them has a unique zero ζ_y in I_j; let θ^c be the zero of a_c and

$$\Gamma(y) := G_y(\zeta_y) - G_y(\theta_j).$$

For the feed column $\Gamma(a) = 0$, since $\zeta = \theta_j$. A mixed column is of type (a) if its distillate is light and its residue heavy, (b) light and mixed, (c) mixed and heavy, (d) mixed and mixed; in type (d), j is an interior root and $\Lambda_{c,T}(\theta^c) = \Lambda_{c,B}(\theta^c) = e_c := V_T - Y_T \geq 0$.

Lemma 17 (local inequality). Under (N), let c be a mixed column, y_1 and y_2 the fluxes of its links to its mixed distillate and residue (when these are mixed), κ the condenser duty on the first of these links (if any), and $\ell_c = A_c$ if c is of type (a) or (b), $\ell_c = 0$ otherwise. Then $\ell_c + \kappa + \sum_e \Gamma(y_e) \geq \Gamma(a_c)$.

Proof. Let $G = G_{a_c} = G^T + G^B$, with G^T, G^B the light parts of $\Lambda_{c,T}$ and $-\Lambda_{c,B}$, and $s_c = \Lambda_{c,T}(\theta^c) = \Lambda_{c,B}(\theta^c) \geq 0$ (Lemma 8). (a) The top section carries light components only: $\Lambda_{c,T} = V_T - G$, so $\ell_c = s_c + G(\theta^c) - G(\theta_j) = s_c + \Gamma(a_c)$. (b) The top carries only light components, the bottom all heavy ones. With $z = \zeta_{y_2}$ ($\Lambda_{c,B}(z) = \rho$), $\ell_c + \Gamma(y_2) - \Gamma(a_c) = s_c - [G^B(\theta^c) - G^B(z)]$. If $z \geq \theta^c$ the bracket is ≤ 0. If $z < \theta^c$, $s_c - \rho = \Lambda_{c,B}(\theta^c) - \Lambda_{c,B}(z) = [G^B(\theta^c) - G^B(z)] + [K^B(z) - K^B(\theta^c)]$ and the second bracket is ≥ 0. (c) Symmetric: with $z = \zeta_{y_1}$, $\kappa + \Gamma(y_1) - \Gamma(a_c) = \kappa + G(z) - G(\theta^c)$; if $z < \theta^c$, $\kappa - s_c = \Lambda_{c,T}(z) - \Lambda_{c,T}(\theta^c) \geq G(\theta^c) - G(z)$. (d) With $e = e_c$, $z_1 = \zeta_{y_1}$ ($\Lambda_{c,T}(z_1) = \kappa$), $z_2 = \zeta_{y_2}$ ($\Lambda_{c,B}(z_2) = \rho$), the claim reads $X := [G^T(z_1) - G^T(\theta^c)] + \kappa - [G^B(\theta^c) - G^B(z_2)] \geq$

0. If $\kappa \le e$ and $\rho \le e$, then $z_1 \ge \theta^c \ge z_2$ and

$$e-\kappa=(z_1-\theta^c)\sum_p x_p,\quad x_p=\frac{\alpha_p d_p}{|\alpha_p-\theta^c||\alpha_p-z_1|},$$
$$e-\rho=(\theta^c-z_2)\sum_p y_p,\quad y_p=\frac{\alpha_p b_p}{|\alpha_p-\theta^c||\alpha_p-z_2|},$$

$G^T(z_1)-G^T(\theta^c)=\sigma_1(e-\kappa)$ and $G^B(\theta^c)-G^B(z_2)=\sigma_2(e-\rho)$, where $\sigma_1=X_L/(X_L+X_H)$, $\sigma_2=Y_L/(Y_L+Y_H)$, $X_L=\sum_{p\le j}x_p$, $X_H=\sum_{p>j}x_p$, and similarly for Y. For $p\le j<q$, $x_py_q-x_qy_p$ is a positive multiple of $d_pb_q\,v_p^1v_q^2-d_qb_p\,v_q^1v_p^2$ with $v_s^t=1/|\alpha_s-z_t|$; since $v_p^1/v_p^2=(\alpha_p-z_2)/(\alpha_p-z_1)\ge 1\ge(z_2-\alpha_q)/(z_1-\alpha_q)=v_q^1/v_q^2$ and $d_pb_q\ge d_qb_p$ (Lemma 16), it is ≥ 0; summing, $X_LY_H\ge X_HY_L$, i.e. $\sigma_1\ge\sigma_2$. Hence $X=\sigma_1e+(1-\sigma_1)\kappa-\sigma_2(e-\rho)\ge(\sigma_1-\sigma_2)e\ge 0$. If $\kappa>e$: $z_1<\theta^c$ and $\kappa-e=\Lambda_{c,T}(z_1)-\Lambda_{c,T}(\theta^c)\ge G^T(\theta^c)-G^T(z_1)$, so the first two terms of X add up to at least e, while the last bracket is $\le e-\rho\le e$ if $\rho\le e$ and ≤ 0 if $\rho>e$. If $\kappa\le e<\rho$: the first bracket is ≥ 0 and the last ≤ 0. □

Theorem 18 (merge-free mixed networks; Agent G)**.** In the permissive model, under (N), if no mixed column of a regular-column configuration has two producers (no *merge node* in the mixed sub-network for j), then $R\ge H_j$.

Proof. The mixed columns form a tree rooted at the feed column; each non-root mixed column c' has a single link e, and $a_{c'}=y_e$. Let $L(c)$ be the sum of ℓ over the mixed columns of the subtree of c plus the condenser duties on the links inside it. From the leaves up, Lemma 17 and $L(c')\ge\Gamma(a_{c'})=\Gamma(y_e)$ give $L(c)\ge\Gamma(a_c)$. At the root, $\sum_{\text{light exits}}\Lambda_s(\theta_j)+Q(\text{mixed})=L([1,N])\ge 0$; conclude with (7). □

For regular-column configurations satisfying (N), Theorem 18 contains Lemma 14 and Lemma 22. Agent G also shows that $R\ge H_j$ holds at every permissive point at which no merge node descends from a type-(d) column whose excess satisfies $\Lambda_{c,T}(\theta_j)>0$ and $\Lambda_{c,B}(\theta_j)>0$ (possible only if $e_c>0$), by combining the sign rule (with superadditivity of $\min(0,\cdot)$ at merges) above those columns and Lemma 17 below them (`general/general-N.md`, Theorem G6). The mechanism that defeats a general proof: Γ is not superadditive at merge nodes (mixing a vapor-rich and a liquid-rich feed lowers the threshold vapor).

6.3 A five-component counterexample in the permissive model

This subsection was added by the independent verifier. The bound $R\ge V_{\mathrm{FTC}}$ is false in the permissive model for $N=5$. Take the regular-column configuration (basic configuration 161 of the Shah–Agrawal space, all transfers by thermal coupling, B, C and D withdrawn as liquid side draws)

$$\begin{aligned}&\mathrm{ABCDE}\to\mathrm{ABCD/BCDE},\quad \mathrm{ABCD}\to\mathrm{ABC/BCD},\quad \mathrm{BCDE}\to\mathrm{BCD/CDE},\\&\mathrm{ABC}\to\mathrm{AB/BC},\quad \mathrm{BCD}\to\mathrm{BC/D},\quad \mathrm{CDE}\to\mathrm{C/DE},\\&\mathrm{AB}\to\mathrm{A/B},\quad \mathrm{BC}\to\mathrm{B/C},\quad \mathrm{DE}\to\mathrm{D/E},\end{aligned}$$

a saturated-vapor feed ($q=0$),

$$\alpha=\left(\tfrac{10675204}{742583},\tfrac{5824255}{854876},\tfrac{529647}{171259},\tfrac{1673613}{789461},1\right)\approx(14.3758,\,6.8130,\,3.0927,\,2.1199,\,1),$$

and the rational feed $z\approx(0.19036,0.18832,0.17089,0.03888,0.41156)$ chosen so that the feed roots are the rationals $\theta\approx(12.3250,5.5538,2.5300,1.9504)$ (exact values in

the file `permissive_counterexample_N5.json` of the folder `verification/`). Then $H \approx (0.33442, 0.32912, 0.46994, 0.43302)$ and $V_{\mathrm{FTC}} = H_3 \approx 0.469941$, while the permissive model has a feasible point with $R \approx 0.469685$, i.e. $R/V_{\mathrm{FTC}} \approx 0.99946$, at which every net flow is at least 10^{-6} (so assumption (N) holds) and every inequality of (3) holds with slack at least 10^{-6}. That an exact feasible point exists within 10^{-30} of the computed one is proved by a Krawczyk test in interval arithmetic on a square subsystem of the equalities (`verification/krawczyk_certificate.py`); the inequalities and $R < V_{\mathrm{FTC}}$ are then checked over the whole box in interval arithmetic.

The mechanism is the one isolated in Section 6.2: for $j = 3$ the column BCDE is of type (d) and runs about 0.006 above its threshold ($V_T \approx 0.03015$, $Y_T \approx 0.02420$) although C and D distribute in it with positive flows, so its root at $j = 3$ is active and an infinite-stage column could not carry this excess; its distillate feeds the merge node BCD, which is not the bottom of a simple diamond. In the strict model, Theorem 11 excludes such a point, and a multistart minimization of the same configuration at the same data gives $R/V_{\mathrm{FTC}} = 1$ to 10^{-7}. Hence the permissive (TG-literal) model is not a faithful model for the conjecture, and of the 20 open (basic configuration, j) pairs at least one, (161, 3), is settled negatively.

Physical meaning. We do not know an adiabatic arrangement with pure products that realizes this point, and the natural candidates fail. (i) An infinite-stage column in which C and D appear in both products has an active root and runs at its Underwood vapor (Halvorsen's rule quoted in Section 2.2); excess vapor at a fixed product distribution requires fewer stages, but a section with finitely many stages cannot remove B from the residue or E from the distillate completely, and as the impurities tend to zero both sections become infinite and the excess is forced to zero. So the point is at best the limit of finite-stage operations with *impure* intermediate streams, which the conjecture does not compare. (ii) A vapor bypass carrying the excess from the node below BCDE to the node above it, or a separate section with $V = L$ between these nodes, transports components (E upward into the node BCD, B downward, or C and D in opposite directions), whereas the permissive excess leaves every net flow unchanged; it therefore realizes a different point, outside Definition 2. The excess-vapor device of TG is physically sound for columns in which at most one component distributes (then V above the threshold does not change the distribution), and the strict model keeps exactly that freedom. We therefore regard the point as an artifact of the permissive formulation, not as a counterexample to a physical reading of the conjecture; it does, however, show that a global optimizer applied to TG's model can return configurations below the FTC value.

6.4 The FTC configuration, every *N*

Theorem 19 (minimum vapor of the generalized Petlyuk arrangement)**.** In both models, every feasible point of the FTC configuration satisfies $R \geq \max_j H_j$.

Proof. Fix j. The mixed streams form the grid (i, m), $1 \leq i \leq j < m \leq N$; the column (i, m) splits $[i, m]$ into $[i, m-1]$ and $[i+1, m]$. Row i is the chain $(i, N) \to (i, N-1) \to \cdots \to (i, j+1)$ of distillate links, ending in the light exit of $(i, j+1)$; the column (i, m) also receives the flux $x_m^{(i)} := -B_{(i-1,m)}$ from row $i-1$ ($x^{(1)} \equiv 0$; all links are couplings or side draws, so (10) has no duty terms). Put $y_m^{(i)} = -B_{(i,m)}$ and $Y_m^{(i)} = \sum_{m' \geq m} y_{m'}^{(i)}$,

$X_m^{(i)} = \sum_{m' \geq m} x_{m'}^{(i)} = Y_m^{(i-1)}$.

Claim: if $X_m^{(i)} \leq 0$ for all m, then $Y_m^{(i)} \leq 0$ for all m. Along the row, $a_{(i,N)} = x_N$ and $a_{(i,m)} = x_m + A_{(i,m+1)}$. We show $A_{(i,m)} \geq X_m$ by descending induction: $A_{(i,N)} \geq \min(0, x_N) = X_N$; if $A_{(i,m+1)} \geq X_{m+1}$ then $a_{(i,m)} \geq X_m$ and $A_{(i,m)} \geq \min(0, a_{(i,m)}) \geq \min(0, X_m) = X_m$ (Lemma 8). Since $y_m = a_{(i,m)} - A_{(i,m)}$, telescoping gives $Y_m = X_m - A_{(i,m)} \leq 0$.

Row 1 has $X \equiv 0$ (the feed column has $a = 0$). By induction $Y_{j+1}^{(j)} \leq 0$; but $-Y_{j+1}^{(j)}$ is the sum of Λ over the heavy exits (the bottom sections of the columns (j, m)). The second form of (7) gives $R - H_j \geq -Y_{j+1}^{(j)} \geq 0$. □

This proves, in the Underwood model, the statement of [1] §3.1 and §5 ("the minimum-energy solution for a generalized Petlyuk arrangement ... is given by the highest peak in the $V_{\min}$ diagram") for every N, the attainment being Theorem 24. When a grid node is missing, the rows no longer absorb the residue fluxes of the previous row, and the induction breaks; family K is the smallest instance.

6.5 Four components

For $N = 4$, $j \in \{1, 3\}$ is Theorem 15. For $j = 2$ the mixed streams are among ABCD, ABC, BCD, BC (A, B, C, D = components 1–4). ABC is present iff it is the distillate of ABCD, BCD iff it is its residue, and BC, if present, is the residue of ABC and the distillate of BCD whenever these are present. Hence: if ABC or BCD is absent, the mixed streams form a path (Lemma 14); if ABC, BCD and BC are present, Lemma 20; the remaining family is

$$\begin{aligned} \mathrm{K} = \{&\mathrm{ABCD} \to \mathrm{ABC/BCD},\ \mathrm{ABC} \to \mathrm{AB/C},\ \mathrm{BCD} \to \mathrm{B/CD}, \\ &\mathrm{AB} \to \mathrm{A/B},\ \mathrm{CD} \to \mathrm{C/D}\} \end{aligned}$$

with its 16 coupling/exchanger variants (B and C are liquid side draws).

Lemma 20 (diamond)**.** If ABC, BCD and BC are present ($N = 4$, $j = 2$), then $R \geq H_2$ in both models.

Proof. Columns 0 = ABCD, 1 = ABC, 2 = BCD, 3 = BC; light exits: tops of 1 and 3. $a_0 = 0$ gives $A_0 = B_0 \geq 0$; $a_1 = A_0 - \kappa_1 \geq -\kappa_1$, $a_2 = -B_0 + \rho_2$, $a_3 = -B_1 + A_2$ (side draw). By Lemma 8, $A_1 \geq \min(0, a_1) \geq -\kappa_1$. If $a_3 \geq 0$ then $A_3 \geq 0$ and $A_1 + A_3 \geq -\kappa_1$. Otherwise $A_3 \geq a_3$ and $A_1 + A_3 \geq A_1 - B_1 + A_2 = a_1 + A_2$; if $a_2 \geq 0$ then $A_2 \geq 0$ and the sum is $\geq -\kappa_1$; if $a_2 < 0$ then $A_2 \geq a_2$ and the sum is $\geq A_0 - \kappa_1 - B_0 + \rho_2 \geq -\kappa_1$. Conclude with (7), $Q(\text{mixed}) = \kappa_1$. □

Lemma 21 (recovery ordering)**.** If the feed column splits ABCD into ABC/BCD (permissive model), its top recoveries satisfy $d_B/z_B > d_C/z_C$. In particular $d_B > 0$ and $b_C = z_C - d_C > 0$.

Proof. The roots of the feed column are $\theta_1, \theta_2, \theta_3$, θ_2 interior. By (3), $\Phi_1 := g_T(\theta_2) - g_T(\theta_1) \geq 0$ and $\Phi_3 := g_T(\theta_2) - g_T(\theta_3) \geq 0$, with $d = (z_A, d_B, d_C, 0)$. On the diagonal $d_B = r z_B$, $d_C = r z_C$ $(0 \leq r \leq 1)$, the feed equation gives $g_T(\theta_t) = r(1 - q) + (1 - r)\mathcal{A}(\theta_t) + r\mathcal{D}(\theta_t)$ with $\mathcal{A}(x) = \alpha_A z_A/(\alpha_A - x)$ increasing and $\mathcal{D}(x) = \alpha_D z_D/(x - \alpha_D)$ decreasing on (α_D, α_A). Thus $\Phi_1 = -(1-r)A_{12} + rD_{12}$ and $\Phi_3 = (1-r)A_{32} - rD_{32}$ with $A_{12} = \mathcal{A}(\theta_1) - \mathcal{A}(\theta_2)$, $D_{12} = \mathcal{D}(\theta_2) - \mathcal{D}(\theta_1)$, $A_{32} = \mathcal{A}(\theta_2) - \mathcal{A}(\theta_3)$, $D_{32} = \mathcal{D}(\theta_3) - \mathcal{D}(\theta_2)$,

all positive, and

$$\frac{A_{12}D_{32}}{A_{32}D_{12}} = \frac{(\alpha_A - \theta_3)(\theta_1 - \alpha_D)}{(\alpha_A - \theta_1)(\theta_3 - \alpha_D)} > 1.$$

For $r = 0$, $\Phi_1 < 0$; for $r = 1$, $\Phi_3 < 0$; for $0 < r < 1$, $\Phi_1, \Phi_3 \geq 0$ would give $A_{12}D_{32} \leq A_{32}D_{12}$. So no feasible point lies on the diagonal. Moreover $\partial\Phi_1/\partial d_B = \alpha_B/(\alpha_B - \theta_2) + \alpha_B/(\theta_1 - \alpha_B) > 0$ and $\partial\Phi_3/\partial d_B = \alpha_B/(\alpha_B - \theta_2) - \alpha_B/(\alpha_B - \theta_3) > 0$: lowering d_B below $z_B d_C/z_C$ lowers both, so $d_B/z_B \leq d_C/z_C$ is infeasible. As the inequality is strict, $d_B > 0$ and $d_C < z_C$. □

Lemma 22 (family K). For every variant of family K, $R \geq H_2$ in both models.

Proof. Columns 0 = ABCD, 1 = ABC, 2 = BCD; light exits: tops of 1 and 2. Since θ_2 is an interior root of column 0, $A_0 = B_0 = V_T - Y_T =: e \geq 0$. Column 1 is fed with $d = (z_A, d_B, d_C)$ and $\Lambda_{1,F} = \Lambda_{0,T} - \kappa_1$; its split AB/C has the single root $\rho_1 := \theta_{1,2} \in I_2$, with $\Lambda_{1,T}(\rho_1) \geq 0$. Column 2 is fed with $b = (b_B, b_C, z_D)$, $\Lambda_{2,F} = -\Lambda_{0,B} + \rho_2'$ (ρ_2' the reboiler duty on the link, 0 for a coupling), root $\rho_2 := \theta_{2,2}$, $\Lambda_{2,T}(\rho_2) \geq 0$. Let $\Delta_p(x) = 1/(\alpha_p - x) - 1/(\alpha_p - \theta_2)$, whose sign is that of $x - \theta_2$ for every p. Then

$$A_1 \geq \sum_{p=A,B} \alpha_p d_p \Delta_p(\rho_1), \quad e - \kappa_1 = \sum_{p=A,B,C} \alpha_p d_p \Delta_p(\rho_1),$$
$$A_2 \geq \alpha_B b_B \Delta_B(\rho_2), \qquad \rho_2' - e = \sum_{p=B,C,D} \alpha_p b_p \Delta_p(\rho_2),$$

the equalities because the feed functions vanish at ρ_1, ρ_2 and equal $e - \kappa_1$, $-e + \rho_2'$ at θ_2. By (7) it suffices to show $A_1 + A_2 + \kappa_1 \geq 0$. If $\rho_1, \rho_2 \geq \theta_2$: $A_1, A_2 \geq 0$. If $\rho_1 \leq \theta_2 \leq \rho_2$: $A_2 \geq 0$ and $A_1 + \kappa_1 \geq e - \alpha_C d_C \Delta_C(\rho_1) \geq e$. If $\rho_1, \rho_2 \leq \theta_2$: $A_1 + \kappa_1 \geq e$ and $A_2 \geq \rho_2' - e - \alpha_C b_C \Delta_C(\rho_2) - \alpha_D z_D \Delta_D(\rho_2) \geq -e$. In the remaining case $\rho_1 \geq \theta_2 \geq \rho_2$ put

$$X = \alpha_C d_C \Delta_C(\rho_1),\ Y = \alpha_B d_B \Delta_B(\rho_1),\ Z = \alpha_B b_B |\Delta_B(\rho_2)|,\ W = \alpha_C b_C |\Delta_C(\rho_2)|,$$

all ≥ 0, with $X + Y \leq e - \kappa_1 \leq e$, $Z + W \leq e - \rho_2' \leq e$ and $A_1 + A_2 + \kappa_1 \geq e - X - Z$. Now $X/(X+Y) + Z/(Z+W) - 1 = (XZ - YW)/((X+Y)(Z+W))$, so $XZ \leq YW$ implies $X + Z \leq e$ (if $X + Y = 0$ or $Z + W = 0$ this is immediate). Finally $XZ \leq YW$ reads

$$\frac{d_C b_B}{d_B b_C} \leq \frac{\mathrm{Rat}(\rho_2)}{\mathrm{Rat}(\rho_1)}, \qquad \mathrm{Rat}(x) := \frac{\Delta_C(x)}{\Delta_B(x)} = \frac{(\alpha_B - x)(\alpha_B - \theta_2)}{(x - \alpha_C)(\theta_2 - \alpha_C)}.$$

The left side is ≤ 1 by Lemma 21 ($d_B z_C \geq d_C z_B \iff d_B b_C \geq d_C b_B$); Rat is decreasing on I_2 and $\rho_1 \geq \rho_2$, so the right side is ≥ 1. (By Lemma 21, $d_B > 0$ and $b_C > 0$, so $\rho_1 < \alpha_B$ and $\rho_2 > \alpha_C$ and all quantities are finite.) □

Theorem 23 ($N = 4$). In both models, every feasible point of every one of the 152 four-component configurations satisfies $R \geq \max(H_1, H_2, H_3)$.

Proof. Theorem 15, Lemmas 14, 20 and 22, by the case distinction above. □

Agent G also proves, for every N and under (N), that $R \geq H_j$ in the permissive model whenever every merge node of the mixed sub-network is the bottom of a *simple diamond*: a type-(d) column whose distillate column is of type (b) and whose residue column is of type (c), both feeding the merge node (`general/general-N.md`, Theorem G8). With

Theorem 18 this gives an independent proof of Theorem 23 under (N); the proof above needs no non-degeneracy assumption.

7 Attainment by the FTC

Theorem 24. For every N, α, $z > 0$ and $q \in \mathbb{R}$, the FTC configuration has an operating point, feasible in the strict (hence in the permissive) model, with $R = \max_j H_j$; all its flows are positive.

Proof. *Preferred split.* Let a column $[i, m]$, $m - i \geq 2$, have positive feed f and net vapor U such that $\theta_i, \dots, \theta_{m-1}$ are roots of its feed equation. Define V_T and d by

$$\alpha_p d_p = V_T \, \frac{\prod_{r=i}^{m-1}(\alpha_p - \theta_r)}{\prod_{p' \in [i,m-1], p' \neq p}(\alpha_p - \alpha_{p'})} \qquad (i \leq p \leq m-1),$$

where the factor V_T is fixed by the condition $d_i = f_i$ (the formula for $p = i$), and $b = f - d$ ($d_m := 0$), $V_B = V_T - U$. Partial fractions give $V_T - g_T(x) = V_T \prod_r (x - \theta_r) / \prod_{p=i}^{m-1}(x - \alpha_p)$, so $g_T(\theta_r) = V_T$ for all r; with the feed equation, $g_B(\theta_r) = V_B$, and since $V_B - g_B$ has its poles at $\alpha_{i+1}, \dots, \alpha_m$, $V_B - g_B(x) = V_B \prod_r (x - \theta_r) / \prod_{p=i+1}^{m}(x - \alpha_p)$ ($V_B \neq 0$, otherwise $b \equiv 0$, contradicting $b_m = f_m > 0$). Counting signs of the factors, every d_p has the sign of V_T and every b_p the sign of V_B; as $d_i = f_i > 0$ and $b_m = f_m > 0$, all are positive. At $x = 0$: $L_T = V_T - \sum d = V_T \prod_r \theta_r / \prod_{p<m} \alpha_p$ and $L_B = V_B + \sum b = V_B \prod_r \theta_r / \prod_{p>i} \alpha_p$, so $0 < L_T < V_T$ and $L_B > V_B > 0$. The column satisfies (3) with $Y_T = V_T$ and all roots active.

Construction. Process the columns by decreasing length. The feed column ($f = z$, $U = 1 - q$) and every column $[i, m]$ with $m - i \geq 2$ are put at the preferred split. A child $[i, m]$ receives d from $[i, m+1]$ and b from $[i-1, m]$ (those present), with $U = V_T[i, m+1] - V_B[i-1, m]$; since both producers have every common root active, $\theta_i, \dots, \theta_{m-1}$ are roots of its feed equation (carry-over [8] [9]), and its feed is positive. A binary column $[r, r+1]$ gets $d_r = f_r$, $b_{r+1} = f_{r+1}$, $V_T = \alpha_r f_r / (\alpha_r - \theta_r) + e_r$, $V_B = \alpha_{r+1} f_{r+1} / (\theta_r - \alpha_{r+1}) + e_r$ with $e_r := \max_s H_s - H_r \geq 0$; it has no interior root, so excess vapor is allowed in both models.

Side draws. It remains to check $V_T[p, p+1] = V_B[p-1, p]$ at the side draws $p = 2, \dots, N-1$ (product purities hold by construction). Let δ_p be the vapor lost at the node of p (inflow from below minus outflow above), treated in Proposition 5 as a condenser duty (or a reboiler duty $-\delta_p$). For the split r, the light exits are the tops of $[i, r+1]$, $i \leq r$; for $i < r$ the column is at its preferred split with root θ_r active, so $\Lambda = 0$; for $i = r$, $\Lambda = e_r$. The only other heat duty is the reboiler of product N. Proposition 5 with $U =$ light streams gives $R - \sum_{p=r+1}^{N-1} \delta_p = H_r + e_r = \max_s H_s$ for $r = 1, \dots, N-1$. For $r = N-1$ this is $R = \max_s H_s$, and successive differences give $\delta_{N-1} = \cdots = \delta_2 = 0$. □

The construction is the one of [1] §3 (every internal column at its preferred split); the excess vapor is placed in the binary columns, whose children are pure products, so that no carried-over root moves.

8 Computer verification and numerical evidence

No proof above depends on a computation. The scripts check the algebra, cross-check the case analysis and search for counterexamples (environment: WSL Fedora, Python 3

with sympy, mpmath, scipy, pyomo, ipopt; paths relative to the problem folder).

- `theory/scripts/check_algebra.py` (sympy, exact): the identities used in Lemmas 21, 22 and Theorem 24. PASS.

- `theory/scripts/check_ftc_achievability.py` (mpmath, 80 digits): builds the point of Theorem 24 for 200 random instances for each $N = 3, \dots, 8$ ($q \in [-1, 2]$) and checks every constraint of the strict model, the side draws and $R = \max H_j$ to 10^{-50}. PASS.

- `theory/scripts/signcheck.py certify4`: decides, by linear programming over the sign patterns of the inflows, which (configuration, j) pairs are not settled by Lemma 8, (10) and the cut identity alone; for $N = 4$ these are exactly the 16 variants of family K at $j = 2$. PASS. (For $N = 5$ the sign rule alone leaves 1312 (configuration, j) pairs for each of $j = 2, 3$, in 2160 of the 6128 configurations and 48 basic configurations. The LPs have integer data but are solved in floating point (HiGHS); the statement that the other pairs are settled by the sign rule is therefore computer-assisted, not a hand proof.)

- Agent C (`computation/`): the enumeration of the configuration space (8/152/6128 configurations, as in [6]); an exact-arithmetic check of the cut identity on all configurations for $N \le 5$; an exact enumeration of sign-consistent light-to-heavy paths (none in the strict model for $N \le 6$; family K only for $N = 4$ in the permissive model); multistart local minimization (ipopt, every accepted point re-verified a posteriori) of R in the strict model over all 152 configurations at 239 random points for $N = 4$, $q = 1$ (minimum ratio $R/V_{\mathrm{FTC}} = 1 - 1.4 \cdot 10^{-9}$), and over all 6128 configurations at the eight test points of [3] for $N = 5$ (minimum $= V_{\mathrm{FTC}}$ to 10^{-10} relative, the FTC attaining it); adversarial joint minimization over (α, z, q) and the operating point (ratio never below 1 beyond 10^{-6}), and targeted minimization on the permissive "bad paths" (family K; the all-coupled variants of the 48 five-component families; 181 of the merge cases), never below 1; SCIP global optimization used only as a cross-check at two points for $N = 4$ (strict model, all 152 configurations, at $\alpha = (8, 4, 2, 1)$, $z_p = 1/4$, $q = 1$: SCIP and ipopt agree to $2.2 \cdot 10^{-9}$, and the SCIP dual bound is $\ge V_{\mathrm{FTC}}$ for every non-tying configuration, while for the tying ones it reached $V_{\mathrm{FTC}}(1 - 8 \cdot 10^{-6})$ at the time limit; permissive model, first 100 configurations, at the point of the example below: agreement to $3.7 \cdot 10^{-9}$). Many exact ties (for $N = 4$, $q = 1$, 12 to 16 configurations tie with the FTC, depending on the dominant peak; 2717 of 2720 observed ties satisfy Corollary 12). Random searches with $q \ne 1$ and a random permissive search for $N = 5$ were not run. The FTC value reproduces the eight five-component optima of [3] (all eight test points have an extreme highest peak, $j = 1$ or $j = 4$, so they are covered by Theorem 15 in both models), and the model reproduces Halvorsen's ternary table [7] (Table 6.1). Scripts and data: `computation/scripts/` and `computation/certificates/`.

- Agent G (`general/scripts/`): `recovery_order_lp.py` (LP minimum of $d_p/f_p - d_{p+1}/f_{p+1}$ over the constraints (3) of random columns, 1587 LPs, positive), `classify_mixed.py` (types and merge nodes of the mixed sub-networks for $N = 4, 5, 6$), `gamma_check.py` (stress test of Lemmas 8, 16, 17 at optimal and random feasible points). `theory/scripts/opencount.py` intersects the sign-hard pairs with the pairs having a merge node ($N = 5$: 864 pairs, 720 configurations; Agent C's path

enumeration gives the same 20 (basic configuration, j) pairs).

- `theory/scripts/stressK.py`: minimizes $d_B/z_B - d_C/z_C$, $A_1 + A_2 + \kappa_1$ and $R - H_2$ over the 16 variants of K in the permissive model: minima $+0.039$, $-2 \cdot 10^{-10}$ and $-2 \cdot 10^{-10}$ (round-off), consistent with Lemmas 21 and 22 and showing that the bound is tight.

- Independent verification (`verification/`, code written from the equations, sharing nothing with the scripts above): an exact (rational) check of the node-balance form of the cut identity on 400 random networks of Definition 2 (several columns per label, merges, heaters and coolers at every node, random subsets U, rational evaluation points); an independent construction of the attainment point of Theorem 24 (60 digits, $N = 3, \dots, 7$, 200 instances); the FTC value at the eight test points of [3]; an independent enumeration (3/8, 18/152, 203/6128) and minimum-vapor NLP for all 8 ternary and all 152 four-component configurations (13 runs in the strict or the permissive model) at 9 parameter points ($q \in \{-0.3, 0.3, 0.4, 0.5, 1, 1.3\}$, four of them with the middle peak highest): Halvorsen's Table 6.1 is reproduced (1.882, 2.032, 2.072, 1.556, 1.366), no configuration falls below V_{FTC} beyond the feasibility tolerance (10^{-7}), and all 120 observed ties satisfy Corollary 12. For $N = 5$, a multistart probe of the 20 open (basic configuration, j) pairs in the permissive model (all-coupled and all-exchanger variants, 220 random parameter points with the peak H_j highest and the others within 10%, $q \in \{-0.3, 0, 0.3, 0.5, 1, 1.2\}$) found one point below V_{FTC}, for the pair (161, 3); after re-optimization with margins it is certified by `krawczyk_certificate.py` (Section 6.3). All other probe values are $\geq 1 - 1.5 \cdot 10^{-7}$.

Example 25 (the identity at work). $N = 4$, $\alpha = (47.5774, 25.5263, 15.1071, 1)$, $z = (0.1358, 0.5169, 0.0923, 0.255)$, $q = 1$ (Agent C, `computation/scripts/cut_identity.py`, local optima of the strict model). The peaks are $H = (0.93322, 1.588798, 0.786044)$, so $V_{\mathrm{FTC}} = H_2 = 1.588798$. The FTC and the all-coupled variant of family K both reach $R = 1.588798$, every term of (7) vanishing at $j = 2$. The K variant with heat exchangers on ABC, BCD and AB has $R = 1.647249$ and $R - H_2 = 0.058451 = 0.021470\,(Q) - 0.002772\,(\Lambda$, top of ABC$) + 0.039753\,(\Lambda$, top of BCD): a negative light-exit flux is compensated. The indirect sequence has $R = 2.481686$. ⌟

9 Scope, limitations and open questions

(1) *Class.* Theorem 11 and Theorem 15 hold for every two-section column network (Definition 2); Theorem 23 is proved for the 152 regular-column configurations with $N = 4$. The wording "any adiabatic distillation arrangement" of [1] would also admit columns with several feed stages (intermediate sections between feeds), stream splitters dividing one section product among several columns, recycles, non-interval intermediate streams, splits in which one product keeps every component of the feed, streams of different labels merged at one feed stage, and reverse net flows; none of these is covered. Heat integration between streams and several pressure levels are excluded by the conjecture itself.

(2) *Reverse flows.* Hypothesis (H1) is part of the model of [3] [4]. The cut identity holds without it, but Lemmas 8 and 9 use the monotonicity of Lemma 4, and the conditions (3) themselves rest on Underwood's interlacing of roots, which fails with reverse flows [7]. Whether reverse flows could lower R is open.

(3) *Model.* The theorems are statements about Underwood's infinite-stage model as formulated by TG, with the split of each column fixed by the network (components $l, \ldots, k$ nominally distributing, possibly with zero flow on one side); its physical justification is that of [7] [3]. At degenerate points (a nominally distributing component absent from one product) the strict model forbids excess vapor that an infinite-stage column could physically carry, and imposes TG's equalities at roots that are not active (Section 2.2); physical operating points of this kind are not covered by Theorem 11.

(4) *Permissive model, $N \geq 5$.* The bound can fail only for splits j whose mixed subnetwork has a merge node below a type-(d) column carrying excess vapor, when the sign rule does not settle the case. For $N = 5$ these are 20 (basic configuration, j) pairs, i.e. 864 (configuration, j) pairs and 720 of the 6128 configurations (in 16 basic configurations), and it does fail for the pair (161, 3) (Section 6.3); the other 19 pairs are undecided (a verifier's multistart search on their all-coupled and all-exchanger variants found no other point below V_{FTC}). All other configurations are covered (Theorems 15 and 18, the latter under (N), and the sign rule, the last by the floating-point LP classification of `signcheck.py`). Numerical searches find no counterexample. Degenerate points (assumption (N) violated) are not covered by Theorem 18.

(5) *Products and feed.* Pure products withdrawn as saturated liquid, a single feed (H2). Nonsharp product specifications and several feeds are not treated. (A pure product withdrawn as saturated vapor instead would only increase the bound: in the node balance a heavy vapor product contributes $\theta_j z_p/(\theta_j - \alpha_p)$ instead of $\alpha_p z_p/(\theta_j - \alpha_p)$, and light products do not enter (6).)

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